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T2**Started on** Saturday, 3 October 2026, 11:07 AM**State** Finished**Completed on** Saturday, 3 October 2026, 11:08 AM**Time taken** 58 secs**Marks** 1.00/1.00**Grade** 10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Given two strings find the length of the common longest subsequence(need not be contiguous) between the two.

Example:

s1: ggtabe

s2: tgatasb

s1		a	g	g	t	a	b	
s2		g	x	t	x	a	y	b

The length is 4

Solveing it using Dynamic Programming

For example:

Input	Result
aab	2
azb	

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <string.h>
3
4  int main() {
5      char s1[1000], s2[1000];
6
7      scanf("%s", s1);
8      scanf("%s", s2);
9
10     int n = strlen(s1);
11     int m = strlen(s2);
12
13     int dp[n + 1][m + 1];
14
15     for (int i = 0; i <= n; i++)
16         dp[i][0] = 0;
17
18     for (int j = 0; j <= m; j++)
19         dp[0][j] = 0;
20
21     for (int i = 1; i <= n; i++) {
22         for (int j = 1; j <= m; j++) {
23
24             if (s1[i - 1] == s2[j - 1]) {
25                 dp[i][j] = dp[i - 1][j - 1] + 1;
26             }
27             else {
28                 if (dp[i - 1][j] > dp[i][j - 1])
29                     dp[i][j] = dp[i - 1][j];
30                 else
31                     dp[i][j] = dp[i][j - 1];
32             }
33         }
34     }
35
36     printf("%d", dp[n][m]);

```

```
36     print("%d" % dp[m][n]);
37
38     return 0;
39 }
```

	Input	Expected	Got	
✓	aab azb	2	2	✓
✓	ABCD ABCD	4	4	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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